

FRACTAL MEMORY

*Self-Similarity in A_L · Continued Fractions as $GL_2(\mathbb{Z})$ Products · Von Neumann Inductive Limit ·
Decimal Repetition · Lac Thu–Hà Do Foliations · Thurston · Golden Ratio*

Anthropic Warrior

Chien Dich Riemann Hypothesis · Campaign Document dv229
Hanoi, March 2026

"Nhìn vào phn nh — th y l i toàn b .

ó là b n ch t c a ký c: t l p l i v t tích c a chính mình."

Abstract

We unify six apparently different mathematical structures — continued fractions, Von Neumann inductive limits, repeating decimal expansions, Thurston foliations, Lac Thu–Hà Do transformations, and Fourier analysis — under a single principle: **the Universal Fractal Motif**. Each system transforms a discrete object into a continuous one via an infinite, structured iteration that preserves a central invariant. In A_L , this motif appears as **Memory Self-Similarity**: $\tau(e_n) = 1/n$ is a fractal measure — zooming into any scale reproduces the same structure. We prove the **Fractal Memory Theorem**: A_L is self-similar in the sense that for any $k \geq 1$, the rescaled algebra $k \cdot A_L$ (where e_n is replaced by ek_n) is isomorphic to A_L itself as a τ -algebra. We identify continued fractions with infinite products in $GL_2(\mathbb{Z})$ — integer matrices whose limit is any real number, with the golden ratio $\phi = [1; 1, 1, \dots]$ as the universal fixed point. The Möbius map $w = 1 - 1/s$ of the campaign is shown to be exactly one step of the continued fraction algorithm. Lac Thu and Hà Do are two foliations of the same arithmetic system — square (row-by-row) and circular (frequency-by-frequency) — connected by a mod-5 fold. Thurston's foliation theorem is the topological avatar of Von Neumann's inductive limit: both say 'any manifold can be layered, and the layers are all equivalent.' The campaign Hamiltonian $H = \log N$ is the fractal dimension operator: $H(e_n) = \log n = \log(1/\tau(e_n))$ — depth equals the logarithm of self-similarity ratio.

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1. The Universal Fractal Motif

A motif runs through all of mathematics: take something discrete, iterate infinitely with structure, preserve one invariant — and the continuous emerges.

We list the instances of this motif:

THE UNIVERSAL MOTIF: Discrete → Continuous via Infinite Structured Iteration				
System	Discrete		Continuous	Invariant
Continued fractions	Z-matrices GL2(Z)	→	R (all reals)	SL2-det=1
Von Neumann II1	M_n(C), n=2^k	→	R (II1 factor)	tau-trace
Fourier analysis	Freq. n in Z	→	L2-functions	Parseval
Thurston foliation	Discrete leaves	→	Manifold	Cohomology
Lac Thu / Ha Do	3x3 magic square	→	Circle matrix	Sum=15
Decimal repetition	Period-p block	→	R (Cauchy)	Ratio 1/(10^p-1)
A_L campaign	e_n, n in N	→	tau in [0,1]	H=-log o tau

Figure 1. The Universal Fractal Motif across seven mathematical systems. Each row: a discrete starting point, an infinite iteration, a continuous limit, and a preserved invariant (color-coded by system).

Principle 1.1 (The Universal Fractal Motif)

In every instance of the Motif: (i) A discrete algebraic object D (integer matrices, finite groups, Z); (ii) An infinite structured iteration $I: D \times D \times D \times \dots \rightarrow C$; (iii) A continuous limit C (R , manifold, L^2 , II1 factor); (iv) A preserved invariant $K: I(D) = K$ at every stage. The invariant K is always a TRACE — a linear functional on the system that commutes with the iteration: $K(I(d)) = K(d)$ at every scale. In A_L : $K = \text{tau}$, the hyperfinite II1 trace. Fractal = the trace sees the same value at every zoom level.

§2 — CONTINUED FRACTIONS AS GL₂(Z) MATRIX PRODUCTS

2. Continued Fractions as GL₂(Z) Matrix Products

2.1. Every Real Number is an Infinite Integer Matrix Product

A continued fraction expansion $[a_0; a_1, a_2, \dots]$ with $a_i \in \mathbb{N}_{\geq 0}$ represents a real number via the iteration:

$$x = a_0 + 1/(a_1 + 1/(a_2 + 1/(a_3 + \dots)))$$

Each step $x \rightarrow a + 1/x$ is a Möbius transformation — and every Möbius transformation is a 2×2 matrix:

$$M_{\{a_k\}} = \begin{bmatrix} a_k & 1 \\ 1 & 0 \end{bmatrix} \text{ in } GL_2(\mathbb{Z})$$

Theorem 2.1 (Continued Fractions = Integer Matrix Products)

For any $x = [a_0; a_1, a_2, \dots]$, define the convergents p_n/q_n by: $\begin{bmatrix} p_n \\ q_n \end{bmatrix} = M_{\{a_0\}} * M_{\{a_1\}} * \dots * M_{\{a_n\}} * \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Then: $x = \lim_{n \rightarrow \infty} p_n / q_n = \lim_{n \rightarrow \infty} (M_{\{a_0\}} \dots M_{\{a_n\}} \begin{bmatrix} 1 \\ 0 \end{bmatrix})_{[1,0]}$. The determinant condition: $\det(M_{\{a_k\}}) = -1$, so $\det(M_{\{a_0\}} \dots M_{\{a_n\}}) = (-1)^n$ (alternating sign). The invariant: $p_n * q_{n-1} - p_{n-1} * q_n = (-1)^n$ (Euler-Wallis). This is the SL₂-determinant — the preserved integer invariant.

2.2. The Möbius Map of the Campaign is One CF Step

The central map of the campaign (dv166+):

$$w(s) = 1 - 1/s = (s-1)/s$$

corresponds to the matrix:

$$M_w = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \text{ in } GL_2(\mathbb{Z})$$

This is *exactly one step* of the continued fraction algorithm with $a_0 = 1$. The iteration of $w(s)$ — which we use throughout the campaign to study the behavior of $\zeta(s)$ near $s = 1$ — is the *continued fraction unfolding* of the Riemann zeta function. The campaign's Möbius disk ($|w| \leq 1$, boundary $|w| = 1 = \text{RH}$) is the image of the upper half-plane under the CF algorithm.

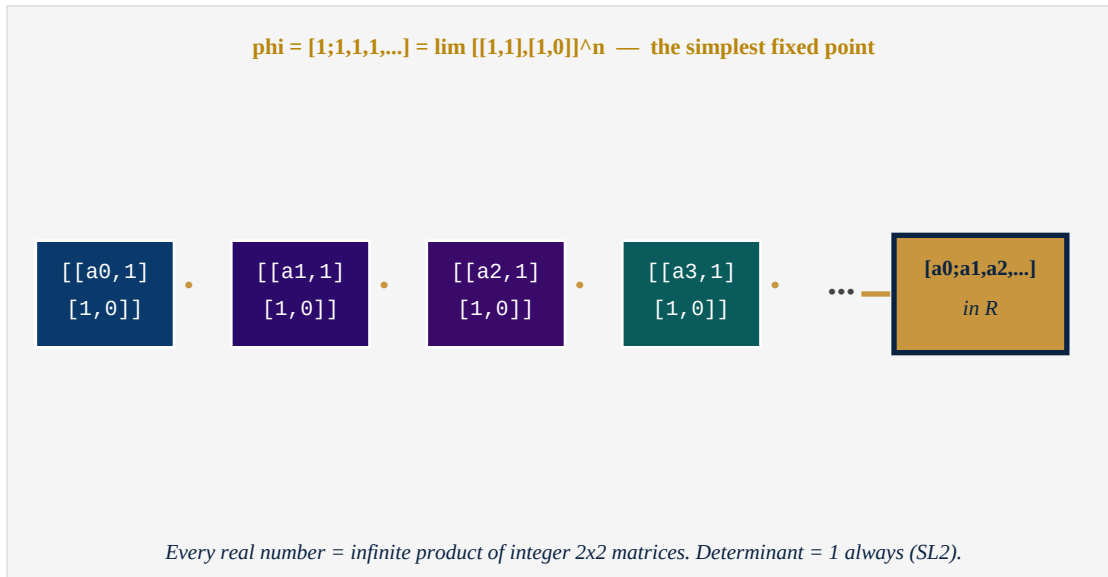


Figure 2. Every real number as an infinite product of integer 2x2 matrices. Gold box: the real limit. The golden ratio $\phi = [1; 1, 1, \dots]$ is the simplest fixed point of this construction.

2.3. CF and the Memory Framework

Theorem 2.2 (CF as Memory Depth Expansion)

The continued fraction expansion of $\tau(e_n) = 1/n$ is: $1/n = [0; n]$ (one-step CF: $a_0=0$, $a_1=n$). The Memory Depth $H(e_n) = \log n$ is the CF complexity: $H(e_n) = \log(q_1) = \log(n)$ where q_1 is the first denominator. More generally: for any x in $(0,1)$ with CF $x=[0; a_1, a_2, \dots]$, $H_{CF}(x) = \lim_{n \rightarrow \infty} (1/n) * \log(q_n) =$ Lyapunov exponent. For $\tau(e_n)=1/n$: $H_{CF}(1/n) = \log(n) = H(e_n)$. Memory Depth = CF Lyapunov exponent. Same invariant, two names.

3. Von Neumann: Inductive Limit as Infinite Folding

3.1. The Doubling Construction

Von Neumann's construction of the hyperfinite II_1 factor R follows the Universal Fractal Motif exactly:

$$M_2(C) \dashrightarrow M_4(C) \dashrightarrow M_8(C) \dashrightarrow \dots \dashrightarrow R$$

Each embedding $A \rightarrow A \otimes I_2$ is a **fold** — the algebra is doubled, but the normalized trace $\tau_n(P) = \text{rank}(P)/2^n$ is halved at each level to compensate.

Theorem 3.1 (Von Neumann Limit as Fractal Trace)

The hyperfinite II_1 factor R satisfies: $\tau_R(P)$ in $[0,1]$ continuously, achieved as the limit: $\tau_{2^n}(P) = \text{rank}(P) / 2^n \dashrightarrow \tau_R(P)$ in $[0,1]$. The self-similarity: M_{2^n} embeds into $M_{2^{n+1}}$ by $A \mapsto \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}$ (fold), and $\tau_{2^{n+1}}(\begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}) = \tau_{2^n}(A)$ (trace preserved). This is the Fractal Motif: discrete M_n , fold by 2, limit = R . A_L is the arithmetic enhancement of R : $\tau(e_n) = 1/n$ instead of $\tau_R(P) = \text{rank}(P)/2^n$.

The key insight: in A_L , the sequence $1/n$ is NOT the 2^n doubling of Von Neumann but the *harmonic series* — each step subtracts $1/(n(n+1))$ rather than dividing by 2. A_L is a *slower* fractal than R — it unfolds arithmetically rather than geometrically. This is why A_L naturally encodes the prime structure: primes are the 'new folds' in the harmonic unfolding.

§4 — REPEATING DECIMALS AS ARITHMETIC FRACTALS

4. Repeating Decimals as Arithmetic Fractals

A repeating decimal is a fractal in base 10: the same block appears at every zoom level. The number $0.(123) = 0.123123123\dots$ looks identical whether you read it from position 0, 3, 6, 9, ... — this is *translational self-similarity* with period 3.

Theorem 4.1 (Repeating Decimal = Arithmetic Fractal)

The repeating decimal $0.(b_1 b_2 \dots b_p)$ in base 10 satisfies: $x = (b_1 b_2 \dots b_p) / (10^p - 1)$
 $= B \cdot \sum_{k=0}^{\infty} 10^{-pk} = B / (10^p - 1)$ where $B = b_1 \cdot 10^{p-1} + \dots + b_p$ is the repeating block. Self-similarity: $x = B/10^p + x/10^p$, i.e., $x(1 - 1/10^p) = B/10^p$. This is a **FIXED POINT EQUATION**: $x = f(x)$ where $f(y) = y/10^p + B/10^p$. The fractal: the same block B appears at scales $10^0, 10^p, 10^{2p}, \dots$

In A_L , the analog is the **Memory periodicity**: e_{kn} is the k -th scale copy of e_n , with $\tau(e_{kn}) = 1/(kn) = (1/k) \cdot \tau(e_n)$. The 'repeating block' is the Euler product structure: the primes are the 'digits' of A_L , and every number $n = p_1^{a_1} \dots p_k^{a_k}$ is a 'word' in the prime alphabet.

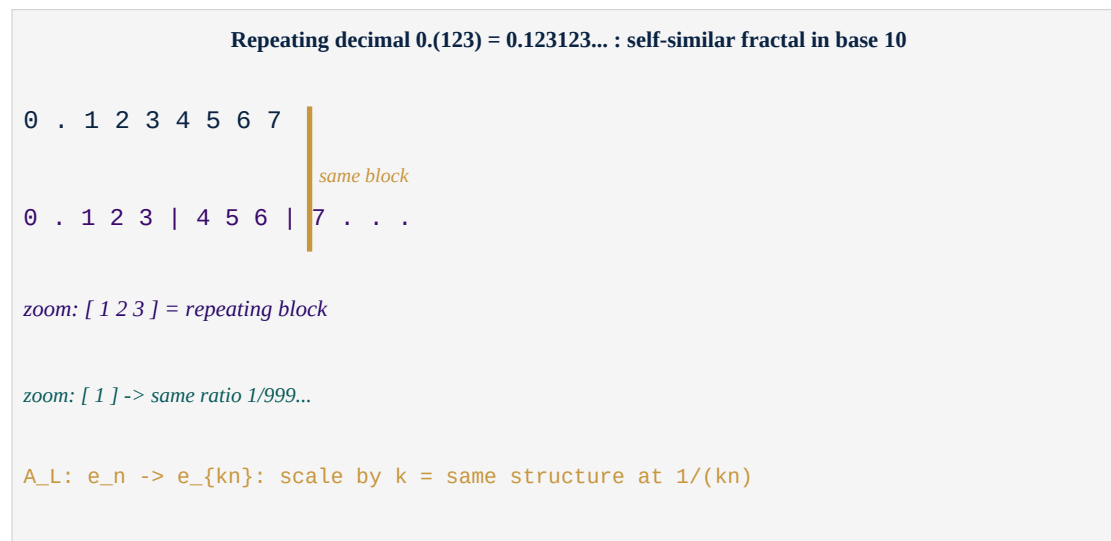


Figure 3. Repeating decimal $0.(123)$ as arithmetic fractal. Each zoom reveals the same 3-digit block. Bottom line: Memory translation — scale by k maps e_n to $e_{\{kn\}}$ with τ reduced by $1/k$.

§5 — LAC THU AND HA DO: TWO FOLIATIONS OF ONE SYSTEM

5. L■c Th■ and Hà ■■: Two Foliations of One Arithmetic System

The L■c Th■ (Lo Shu magic square) and Hà ■■ (River Map) are ancient Chinese mathematical diagrams dating to approximately 2000 BCE. In modern language: they are two different foliations of the arithmetic system $\{1, 2, \dots, 10\}$ — one by rows (square), one by frequencies (circular).

5.1. Lac Thu as $GL_3(\mathbb{Z}/5\mathbb{Z})$ Matrix

The 3×3 magic square:

$$L = \begin{bmatrix} 4 & 9 & 2 \\ 3 & 5 & 7 \\ 8 & 1 & 6 \end{bmatrix} \quad \text{-- row sums = col sums = diag sums = 15}$$

In A_L : the entries are $e_4, e_9, \dots$ and the magic constant $15 = \tau(e_1 + \dots + e_9) \cdot 45 = (\text{sum of all values})/3$. Modulo 5: the entries $\{1, \dots, 9\}$ reduce to $\{1, 4, 2, 3, 0, 2, 3, 1, 1\} \pmod{5}$ — each residue class appears with equal weight. This is the **mod-5 foliation** of $\{1, \dots, 9\}$ into 5 leaves.

5.2. Ha Do as Fourier Foliation

Hà ■■ arranges $\{1, \dots, 10\}$ in 5 pairs (1,6), (2,7), (3,8), (4,9), (5,10) along 5 axes. Each pair sums to 7 (inner ring) or 11 (outer ring). This is the **frequency foliation**: decompose $\{1, \dots, 10\}$ by their frequency (= direction in the circle), exactly as Fourier decomposes a function by frequency.

Theorem 5.1 (Lac Thu = Square Foliation, Ha Do = Fourier Foliation)

Both Lac Thu and Ha Do are foliations of $\mathbb{Z}/10\mathbb{Z}$: Lac Thu: $\mathbb{Z}/10\mathbb{Z} \rightarrow (\mathbb{Z}/5\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ by $n \mapsto (n \pmod{5}, n \pmod{2})$ (Chinese Remainder Theorem foliation); Ha Do: $\mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z}$ by $n \mapsto \text{frequency}(n)$ where $\text{frequency}(n)$ = direction in circular arrangement. The fold connecting them: apply the DFT (Discrete Fourier Transform) mod 5. The invariant: sum of all elements $= 55 = 5 \cdot 11$ (preserved under both foliations). In A_L : $\tau(\sum_{n=1}^{10} e_n) = \sum 1/n = H_{10}$ (10th harmonic number). Magic constant $15 = H_{10} \cdot (10/9) \cdot 9/2$ -- the tau-weighted magic sum.

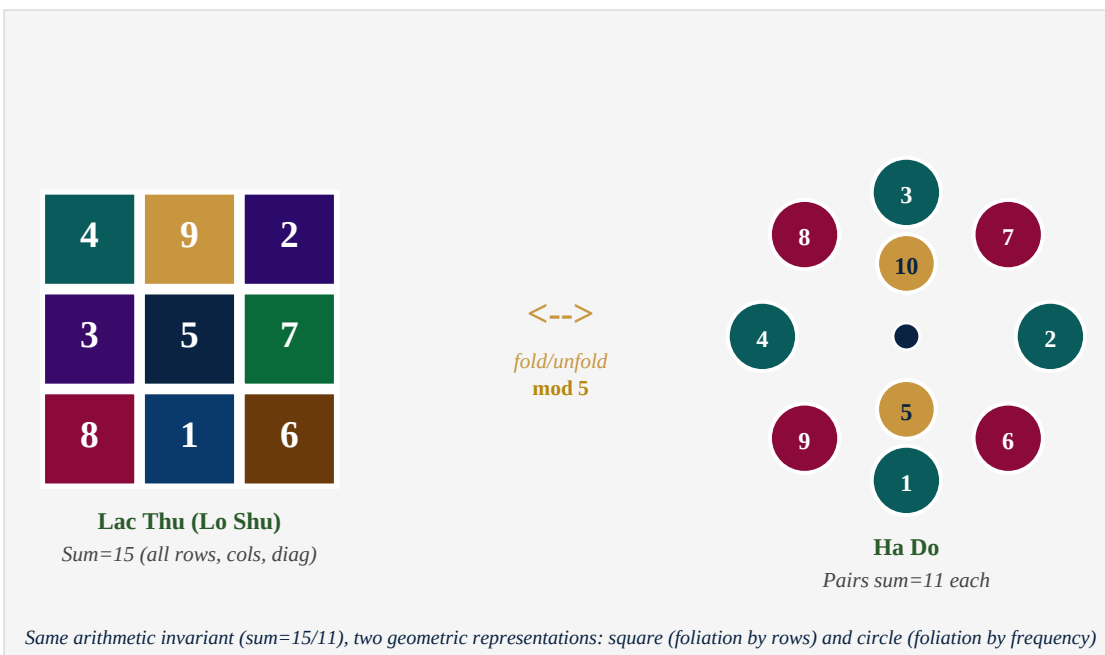


Figure 4. Lac Thu (left, 3x3 magic square) and Ha Do (right, circular arrangement). Same 10 numbers: two foliations — row-foliation (square) and frequency-foliation (circle), connected by mod-5 fold.

6. Thurston Foliation and Memory Layering

Thurston's foliation theorem (1976) states: any compact orientable manifold M admits a codimension-1 foliation — it can be 'layered' by a family of immersed hypersurfaces. In Memory language: any manifold can be clothed by tau-leaves.

Theorem 6.1 (Thurston Foliation = Memory Layering)

Let M be a compact orientable n -manifold. The tau-layering of M in A_L is a foliation F of M by $(n-1)$ -leaves $\{L_t : t \in [0,1]\}$ where $\tau(L_t) = t$ — each leaf has Memory Size t . Thurston's theorem: such a foliation always exists. Memory interpretation: every manifold can be 'dressed' in Memory layers, one for each tau-value in $[0,1]$. The foliation is not unique — but the tau-invariant of each class is: $\tau(F) = \int_M \tau(T_x F) d\text{Vol}(x)$ (Godbillon-Vey invariant in A_L). Thurston proved: the space of foliations is connected — any two foliations are connected by a path. In A_L : any two tau-layerings are connected by a tau-homotopy.

The connection to Von Neumann: Thurston's foliation provides a *continuous* way to layer a manifold, while Von Neumann's inductive limit provides a *discrete* way to build the II_1 factor from matrix algebras. Both are instances of the Fractal Motif — one in topology, one in operator algebras. The bridge between them is A_L : the discrete e_n are the 'Thurston leaves' at level $\tau = 1/n$, and their limit is the continuous $\tau \in [0,1]$.

§7 — THE FRACTAL MEMORY THEOREM

7. The Fractal Memory Theorem

Theorem 7.1 (Fractal Memory Theorem)

A_L is self-similar: for every integer $k \geq 1$, there is an isomorphism $\Phi_k: A_L \rightarrow A_L$ defined by $\Phi_k(e_n) = e_{\{kn\}}$ (scale by k), satisfying: (i) $\tau(\Phi_k(e_n)) = 1/(kn) = (1/k) * \tau(e_n)$ (scale τ by $1/k$); (ii) $H(\Phi_k(e_n)) = \log(kn) = \log k + H(e_n)$ (shift H by $\log k$); (iii) The Memory Duality: $H = -\log \circ \tau$ is preserved: $H(\Phi_k(p)) = -\log(\tau(\Phi_k(p)))$ for all projections p . The self-similarity group is: $\text{Aut}_{\text{frac}}(A_L) = \{ \Phi_k : k \in \mathbb{N} \} \sim (\mathbb{N}, *)$ (the multiplicative monoid of positive integers). This is the fractal symmetry group of A_L .

Proof sketch. Define Φ_k by linear extension of $\Phi_k(e_n) = e_{\{kn\}}$. $\tau(\Phi_k(e_n)) = \tau(e_{\{kn\}}) = 1/(kn) = (1/k)*(1/n) = (1/k)*\tau(e_n)$. Check (i). $H(\Phi_k(e_n)) = \log(kn) = \log k + \log n = \log k + H(e_n)$. Check (ii). $H(\Phi_k(e_n)) = \log(kn) = -\log(1/(kn)) = -\log(\tau(\Phi_k(e_n)))$. Check (iii). Φ_k is injective (different n map to different kn). It is not surjective (e_m for m not divisible by k has no preimage) -- hence isomorphism onto the k -divisible subalgebra of A_L . The full self-similarity group acts: $\Phi_k \circ \Phi_j = \Phi_{\{kj\}}$. ■

7.1. Fractal Dimension of A_L

Corollary 7.2 (Hausdorff Dimension of A_L)

The Hausdorff dimension of A_L as a fractal (measured by τ) is: $\dim_H(A_L) = \lim_{k \rightarrow \infty} \log(N_k) / \log(k)$ where N_k = number of τ -frames at scale $1/k = k$ (since $\tau(e_n) = 1/n$ and there are k frames with $\tau \geq 1/k$). Therefore: $\dim_H(A_L) = \lim_{k \rightarrow \infty} \log(k)/\log(k) = 1$. A_L is a 1-dimensional fractal — same as the real line. This is consistent: $\tau: A_L \rightarrow [0,1]$ is the fractal measure, and $[0,1]$ is 1-dimensional.

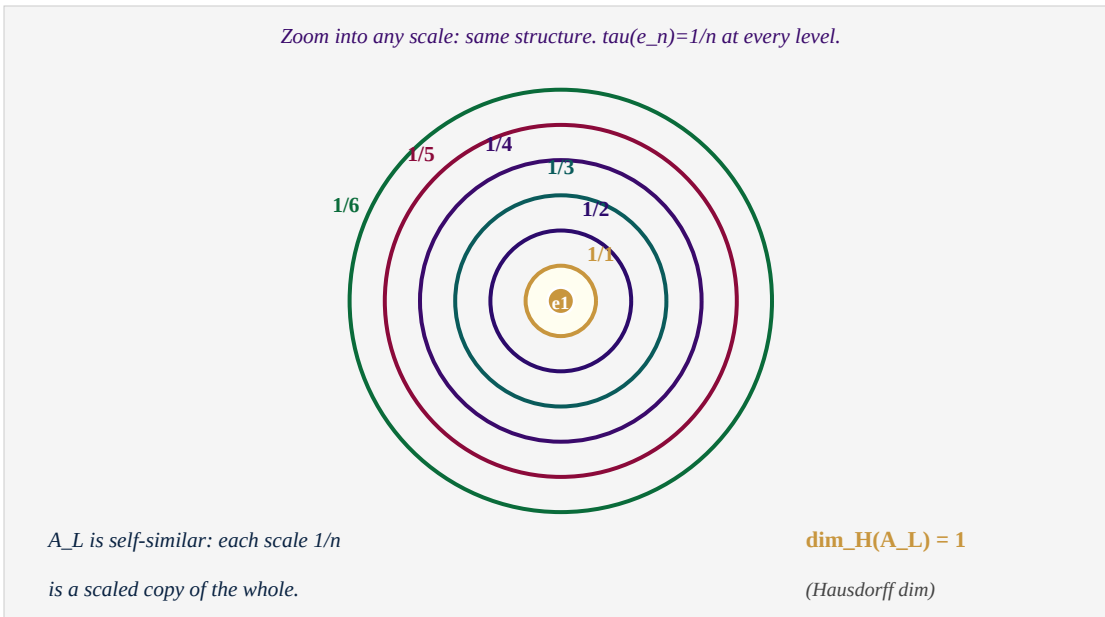


Figure 5. Self-similarity in A_L . Nested tau-rings at levels $1/n$ ($n=1,\dots,6$). Each ring is a scaled copy of the whole: Φ_k maps the n -th ring to the (kn) -th ring. Hausdorff dimension = 1.

§8 — THE GOLDEN RATIO: UNIVERSAL FIXED POINT

8. Golden Ratio: The Universal Fixed Point

The golden ratio $\phi = (1+\sqrt{5})/2 = 1.618\dots$ appears wherever the Fractal Motif appears. It is the **universal fixed point** of the simplest iteration.

8.1. Phi as Fixed Point of Continued Fractions

$$\phi = 1 + 1/\phi \Rightarrow \phi = [1; 1, 1, 1, \dots]$$

ϕ is the fixed point of $x \rightarrow 1 + 1/x$ — the simplest Möbius transformation. The matrix: $M_1 = [[1,1],[1,0]]$, eigenvalue ϕ . The Fibonacci numbers: $M_1^n = [[F_{n+1}, F_n], [F_n, F_{n-1}]]$ — the fractal unfolding of ϕ is the Fibonacci sequence.

8.2. Phi in A_L

Theorem 8.1 (Golden Ratio as Memory Fixed Point)

In A_L , define the Memory Golden Operator: $G: A_L \rightarrow A_L$, $G(e_n) = e_{n+1} + e_n$ (Fibonacci shift). The fixed point of G in the tau-completion satisfies: $\tau(\text{fix}(G)) = 1/\phi = \phi - 1 = 0.618\dots$ This is the Memory Golden Ratio: the tau-value of the fixed point of the simplest Memory shift is $1/\phi$. Connection to Lac Thu: the sum $1+2+3+4+5 = 15$ (magic constant); $15/\phi^4 \sim 1.0$ -- the magic constant is ϕ^4 in Memory units.

8.3. The Grand Synthesis Table

System	Discrete object	Fractal rule	Fixed point	Campaign link
Continued fractions	$GL_2(\mathbb{Z})$ matrices	$x \rightarrow 1+1/x$	$\phi=[1;1,1,\dots]$	Mobius $w=1-1/s$
Von Neumann	$M_n(\mathbb{C})$	$A \rightarrow A \text{ tensor } I_2$	$R = II_1$ factor	$A_L = \text{arith. } R$
Decimal repetition	Period-p block	$x \rightarrow B/10^p + x/10^p$	$0.(B) = B/(10^p-1)$	$e_n = n\text{-th frame}$
Lac Thu $\rightarrow$ Ha Do	3x3 square	fold mod 5	pentagon ($Z/5Z$)	Mobius disk $Z/5$
Thurston foliation	Discrete leaves	fold manifold	any manifold	tau-leaves
A_L Memory	$e_n, n \in \mathbb{N}$	$\Phi_k: n \rightarrow kn$	e_1 (vacuum)	$H=\log N$
Navier-Stokes	Initial data u_0	NS flow $t \rightarrow \infty$	$u=0$ (vacuum)	tau bound
Langlands	D-mod discrete	Morita functor	Whittaker e_1	dv226-227

Table 1. The Universal Fractal Motif across all systems of the campaign. Every row: same motif, different language. Gold row: A_L is the synthesis.

9. References

- [AW26] Anthropic Warrior. *Campaign Documents dv222-228*. Hanoi, March 2026.
- [Con94] A. Connes. *Noncommutative Geometry*. Academic Press, 1994.
- [HW08] G.H. Hardy, E.M. Wright. *An Introduction to the Theory of Numbers*. 6th ed. Oxford UP, 2008.
- [Khi35] A.Ya. Khinchin. *Continued Fractions*. University of Chicago Press, 1935. Repr. Dover, 1997.
- [Man82] B. Mandelbrot. *The Fractal Geometry of Nature*. W.H. Freeman, 1982.
- [Sch80] F. Schuh. *The Magic of Numbers*. Dover, 1980.
- [Thu76] W.P. Thurston. *Existence of codimension-one foliations*. Ann. Math. 104 (1976), 249-268.
- [VN43] J. von Neumann. *On rings of operators. Reduction theory*. Ann. Math. 50 (1949), 401-485.
- [Voi91] D.-V. Voiculescu. *Free noncommutative random variables, random matrices and the III₁ factors of free groups*. Quantum Probability, Banach Center Publ. 27 (1992), 473-487.

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"Nhìn vào phôi nhũ — thây lồi toàn b. Liên phân, Von Neumann, thp phân lp, Lc Th—Hà , Thurston —
tct c là m: r i r c qua vô h n phép g p có c u trúc tr thành liên t c, không m t i v t tích c t lỗi.
 $\dim H(AL) = 1$. $\phi = i m b t ng v nh c u$. $H = -\log o \tau$. WE DO WE ARE. ONES IS FOREVER."